

Pixel Pilot Architecture Overview

1. Purpose

Pixel Pilot is a web-based dashboard for managing, executing, and reporting BackstopJS visual regression tests at scale. It supports batch processing, scenario management, real-time progress, and historical test tracking.

2. High-Level Components

A. Frontend (React)

- UI Components: ScenarioManager, TestRunner, ConfigEditor, ScreenshotUploader, URL Clone Tool, Batch History Viewer.
- State Management: Local React state, API-driven updates.
- Real-Time Updates: Socket.IO for live test progress and notifications.
- CSV Import/Export: For bulk scenario management and feature status tracking.

B. Backend (Node.js/Express)

- API Endpoints: RESTful endpoints for scenario CRUD, batch test execution, config management, batch history, and report serving.
- Batch Processor: Handles large test suites, splits into batches, manages concurrency, and writes results to timestamped folders.
- BackstopJS Integration: Spawns BackstopJS CLI for each batch, with isolated config and output folders.
- File Management: Uses fs-extra for robust file operations, including batch folder creation and metadata storage.
- Socket.IO: Pushes real-time progress, warnings, and completion events to the frontend.

C. Data Storage

- Config Files: BackstopJS config per project, editable via UI.
 - Batch Folders: Each test run (even single scenario) creates a unique batch folder with timestamp, containing all test assets and metadata.
 - Reference Images: Stored in a static folder, can be updated from any batch.
 - Batch Metadata: Each batch folder contains a batch-meta.json file with run details (timestamp, scenario count, status).
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3. Key Workflows

A. Scenario Management

- Add/edit/delete/search scenarios via UI.
- Import/export scenarios as CSV.
- Filter scenarios for targeted test runs.

B. Test Execution

- User selects scenarios and starts a test.
- Frontend sends batch config to backend (always batch, even for single scenario).
- Backend validates only filtered scenarios, creates batch folder, and runs BackstopJS.
- Progress and results streamed to frontend via Socket.IO.

C. Batch History & Reporting

- Each run is preserved in its own batch folder.
- Batch metadata and reports are accessible via API and UI.
- Users can browse, compare, and export results.

D. Reference Management

- Reference images are static.
- Option to update/copy reference images from any batch.

E. URL Cloning & Extraction

- Clone URLs from target/reference sites using CSS selectors.
- Ensures unique scenario labels and proper mapping.

4. API Endpoints (Sample)

- POST /api/projects/:projectId/test — Run batch test.
- GET /api/projects/:projectId/batches — List batch history and metadata.
- GET /api/projects/:projectId/report — Serve batch reports.
- POST /api/projects/:projectId/scenarios — Manage scenarios.
- POST /api/clone-urls — Clone URLs from a site.

5. Batch Folder Structure

```
backstop_data/  
  bitmaps_test/  
    batch_0/  
      20250913-123456/  
        [test images, logs, batch-meta.json, report]  
    batch_1/  
      20250914-101010/  
      ...  
  bitmaps_reference/  
    [reference images]  
  html_report/  
    batch_0/  
      20250913-123456/  
        index.html, report.json  
    ...
```

6. Extensibility

- New features (permissions, CI/CD integration, custom exports) can be added via new API endpoints and UI components.
 - Batch logic is consistent for all runs, enabling robust history and analytics.
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7. Technology Stack

- **Frontend:** React, Material UI, Axios, Socket.IO-client
 - **Backend:** Node.js, Express, Socket.IO, BackstopJS, fs-extra, cheerio, axios
 - **Data:** JSON config, batch metadata, CSV for import/export
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8. Security & Permissions

- (Planned) Role-based access for teams.
 - API endpoints can be secured via middleware.
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9. Deployment

- Can be deployed on any Node.js-compatible server.
 - Frontend served via Vite/React build.
 - Backend runs as Express server.
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10. Architecture Diagram



11. Summary

Pixel Pilot provides a scalable, user-friendly platform for visual regression testing, with robust batch history, real-time feedback, and extensible architecture for enterprise needs.